

Trace and Determinant

Math 5250 — Week 2

Trace

Definition. For an $n \times n$ matrix A over a field F , the **trace** is

$$\text{trace}(A) = \sum_{i=1}^n a_{ii}.$$

Proposition. If A, B are $n \times n$ matrices over F , then $\text{trace}(AB) = \text{trace}(BA)$.

Trace of a Linear Map

Let $f : V \rightarrow V$ be linear, V finite dimensional.

Definition. Choosing a basis E for V , $\text{trace}(f) = \text{trace}([f]_E^E)$.

This is independent of the choice of basis: for a different basis E' ,

$$\text{trace}([f]_{E'}^{E'}) = \text{trace}([f]_E^E).$$

Trace, Subspaces, and Quotient Spaces

Let $L : V \rightarrow V$ be linear and $W \subset V$ an **invariant subspace** ($L(W) \subset W$). Then L induces a well-defined map $\bar{L} : V/W \rightarrow V/W$ by $\bar{L}([v]) = [L(v)]$.

Proposition. With L_W the restriction of L to W ,

$$\text{trace}(L) = \text{trace}(L_W) + \text{trace}(\bar{L}).$$

Trace and Short Exact Sequences

If $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is a short exact sequence and $L = (L_1, L_2, L_3)$ is a compatible linear map $(A, B, C) \rightarrow (A, B, C)$, then

$$\text{trace}(L_1) - \text{trace}(L_2) + \text{trace}(L_3) = 0.$$

More generally, for a bounded complex $X^\cdot$ of finite dimensional spaces and a linear map $L : X^\cdot \rightarrow X^\cdot$,

$$\text{trace}(L|X^\cdot) := \sum_i (-1)^i \text{trace}(L_i) = \sum_i (-1)^i \text{trace}(\bar{L}_i),$$

where $\bar{L}_i$ are the induced maps on cohomology.

Determinant

Proposition. There is a unique function $\det : M_n(F) \rightarrow F$ such that:

1. $\det$ is multilinear in the rows of A .
2. $\det$ is alternating: it vanishes when two rows are repeated (so swapping two rows negates $\det$).
3. $\det(I) = 1$.

Effect of Row Operations on the Determinant

Assuming $\det$ exists as above:

- a. Swapping two rows of A : $\det(A) = -\det(A')$.
- b. Scaling a row by a : $\det(A) = a \det(A')$.
- c. Adding a multiple of one row to another: $\det(A) = \det(A')$.

Existence of the Determinant

- ▶ 1×1 : $\det(a) = a$. 2×2 : $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$.
- ▶ Inductively, with A_i the submatrix deleting row 1 and column i ,

$$\det(A) = \sum_{i=1}^n (-1)^{i+1} a_{1i} \det(A_i).$$

This is the cofactor expansion along the first row (or any row or column).

Determinant is Multiplicative

Proposition. $\det(AB) = \det(A) \det(B)$.

Corollary. A matrix is invertible if and only if its determinant is nonzero.

Permutations and the Determinant

For the standard row vectors e_i , $\det(e_{i_1}, \dots, e_{i_n}) = \pm 1$; the permutation $i_1, \dots, i_n$ is **even** if this is $+1$, **odd** if -1 — equivalently, according to the parity of the number of exchanges needed to sort it.

Expansion.

$$\det(A) = \sum a_{1i_1} a_{2i_2} \cdots a_{ni_n} \det(e_{i_1}, \dots, e_{i_n}),$$

a sum of $n!$ terms (the Leibniz formula).

The Adjugate Matrix

Definition. The **adjugate** A^{adj} has entries $b_{ij} = (-1)^{i+j} \det(A_{ji})$, where A_{ji} deletes row j and column i of A .

Proposition. $AA^{adj} = A^{adj}A = \det(A) I$.

Remark. This gives an explicit formula for the inverse:
 $A^{-1} = \det(A)^{-1} A^{adj}$.

Cramer's Rule

For $Ax = b$ with $\det(A) \neq 0$,

$$x_j = \frac{\det(A_j(b))}{\det(A)},$$

where $A_j(b)$ is A with column j replaced by b .

Remark. If A, b have integer entries and $\det(A) = \pm 1$, then x has integer entries.

Determinant of a Linear Map

Let $f : V \rightarrow V$ be linear, V finite dimensional over F .

Definition. $\det(f) = \det([f]_E^E)$ for any basis E .

This is well-defined: changing basis conjugates the matrix, which does not change the determinant.

Determinants and Short Exact Sequences

Proposition. If $0 \rightarrow A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow 0$ is a short exact sequence of finite dimensional spaces and $L = (L_1, L_2, L_3)$ is a compatible linear map, then

$$\det(L_2) = \det(L_1) \det(L_3).$$

More generally, for a bounded complex $X^\cdot$ and $L : X^\cdot \rightarrow X^\cdot$,

$$\sum_i (-1)^i \det(L^i : X^i \rightarrow X^i) = \sum_i (-1)^i \det(\bar{L}^i : H^i(X^\cdot) \rightarrow H^i(X^\cdot)).$$